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CSA1021 INTRODUCTION TO PROBLEM SOLVING AND PROGRAMMING

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1. Introduction

The Student Management System is a Python-based project developed to manage student information in a simple and organized way. The system allows the user to add, view, search, update and delete student records. The main purpose of this project is to demonstrate the practical use of Python programming concepts such as variables, data types, conditional statements, loops, functions, lists/dictionaries and menu-driven programming.

1. **Objectives**

The main objectives of the Student Management System are: To store student information systematically. To add new student records. To display existing student records. To search for a particular student. To update student information. To delete a student record. To provide a simple and user-friendly menu. To apply basic Python programming concepts in a practical project.

1. **Problem**

Managing student records manually can be time-consuming and may lead to errors. It can also become difficult to find or update a particular student's information. Therefore, the Student Management System

provides a simple computerized solution for maintaining student records efficiently.

1. **Technologies**

Used Technology Purpose Python Programming language
IDLE / VS Code Development environment Lists / Dictionaries
Storing student data Functions Dividing the program into
modules Conditional Statements Decision making Loops
Repeated operations

2. **Main Features**

1 Add Student : The user can enter details such as student ID, name, age, branch and marks.

.2 Display Students: The system displays all the student records stored in the program

.3 Search Student: The user can search for a student using the student ID.

4 Update Student: Existing student information can be modified whenever required..

5 Delete Student: A student record can be removed from the system using the student ID.

6 Exit: The user can safely exit the program through the exit option.

1. **Working of the Project:** The program starts by displaying a menu of available operations. The user selects an option according to the required task. The selected operation is performed using appropriate Python functions. After completing the operation, the menu is displayed again so that the user can perform another task. The program continues until the user selects the Exit option.
2. **Python Concepts Used** The project demonstrates several fundamental Python concepts: Variables and data types Input and output if -elif -else statements while and for loops Functions Lists Dictionaries String operations Menu-driven programming
3. **Advantages** Easy to use. Saves time in managing student records. Reduces manual work. Makes searching student information provides organized record management. Helps beginners understand practical Python programming.
4. **Limitations** The basic version stores data temporarily while the program is running. It does not use a database. It is designed for basic student record management. It does not provide login or authentication features.
5. **Future Scope** The project can be improved in the future by adding: Database connectivity using MySQL or SQLite. Student

login and administrator login. Attendance management. Subject-wise marks and grades. Fee management. Graphical User Interface (GUI). Exporting student records to Excel or PDF. Cloud-based student record storage.

6. **Conclusion**

The Student Management System is a simple Python project designed to manage student records efficiently. It demonstrates the practical implementation of basic programming concepts learned in Python. This project helped in understanding functions, loops, conditional statements, data structures and menu-driven programming. With additional features such as database connectivity and GUI, the system can be developed into a more advanced student management application.